



MINING DEALER BEST PRACTICE

789 FINAL DRIVE INNER CARRIER LIFTING TOOL

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

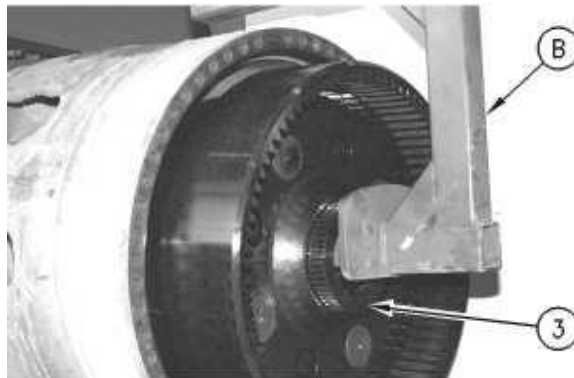
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October 18
4.03-160927-02

1.0 Introduction

Although the horizontal disassembly process is displayed for the 789 OHT Final Drive, no specific information is available on SISWEB about the first carrier's removal procedure. Nonetheless, at RELIANZ, both assembly and disassembly processes are carried out vertically. For this reason, a new tool is necessary to ensure the safe handling of the equipment and the expedition of the task to the technician; finally pointing to the time reduction required and the difficulty of the procedure.



This best practice lays out the design of a tool that will allow the removal of the first carrier while placing the 789 final drive component vertically.

2.0 Best Practice Description

The tool consists of an ASTM A36 "T-shaped" steel block, to which a threaded ring (CROSBY M24X36 4-ton capacity) and a lifting equipment hook are attached. The tool also has two hook-shaped ASTM A36 steel blocks on each side (symmetrical to the vertical axis of the lifting ring) which are assembled onto the block by means of UNC ½" hexagonal threaded bolts. These blocks are the ones that finally hold the carrier.

The main features of the tool are:

Dimensions: 100mm x 230mm x 83mm (D x W x H)

Total weight: 9.30 Kg.



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3.0 Implementation Steps

- Review the status of the tool (1) to verify its integrity before usage.
- Once the final drive's outside carrier is removed, place the lifting tool in the upper opening of the sun gear and adjust it to the required diameter (2).
- Secure the moving parts as shown in Fig (2).
- By using the Jib crane, hoist the carrier (3) to the desired height.
- Locate the component in its final housing (stowage, basket, box) (4) and remove the tools (5).



4.0 Benefits

The main benefits provided by this implementation are:

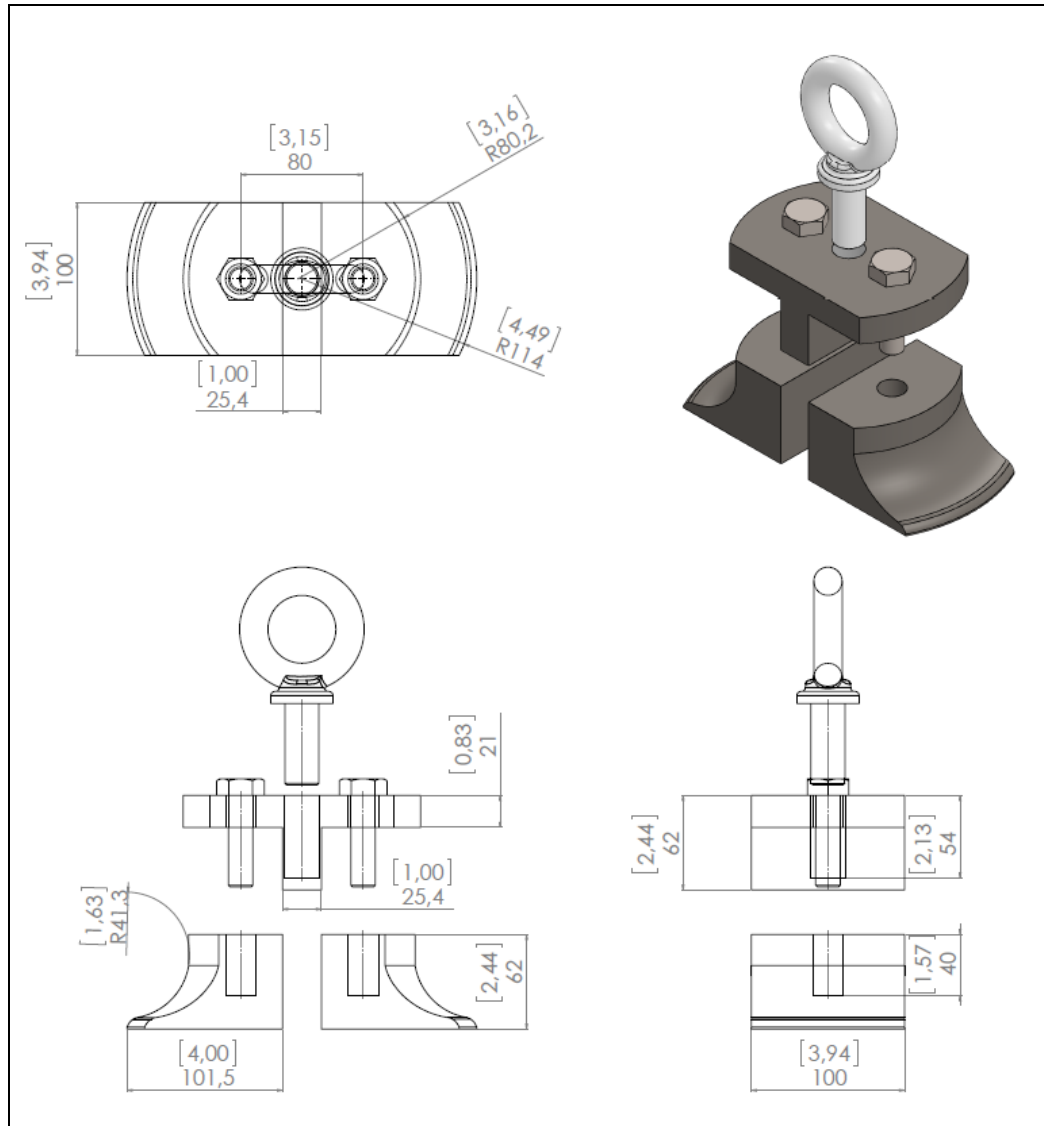
- Reduction in the removal time of the planetary carrier.
- Reduction in the ergonomic risk of the technicians.
- Easiness in the performance of the task.

5.0 Resources Required

Welder, equipment and materials according to the tool drawings.

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6.0 Supporting Attachments / References



Nombre de carga	Cargar imagen	Detalles de carga
Gravedad-1		Referencia: Planta Valores: 0 0-9.81 Unidades: SI
Fuerza-1		Entidades: 4 cara(s) Referencia: Arista< 1 > Tipo: Aplicar fuerza Valores: ---, ---, -2452 N

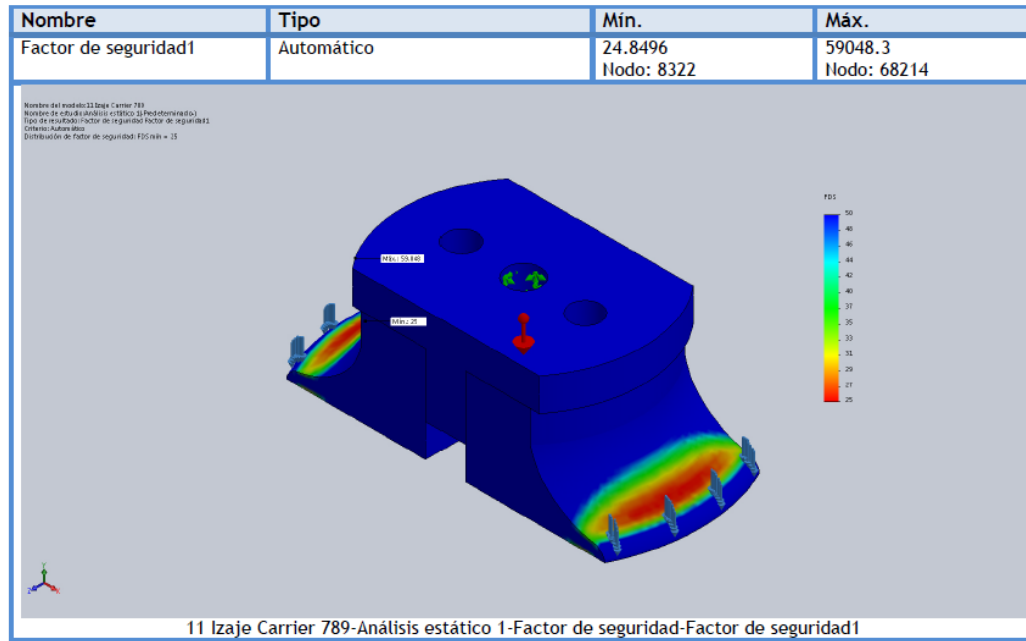
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The minimum safety factor related to this tool is 24.8. Since this safety factor is greater than 5.0, the tool is indeed a safe one to use.

7.0 Related Best Practices

None.

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